

Meeting to discuss the inhibit...for Quake Aim 1 Ruggedization

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SUMMARY KEYWORDS

inhibitory neuron, synaptic assay, GABA release, chloride reversal, excitability assay, single-cell stimulation, GABA receptors, synaptic machinery, GABA sniffer, GABA B receptors, GABA A receptors, neuronal maturation, synaptic cleft, action potential, synaptic transmission

SPEAKERS

Speaker 3, Steven Ryan, Speaker 1, Aishwarya Dhandapani, Speaker 2, James Fink



00:03

Coming. I'm coming over. I



00:30

see, she says, Tik Tok, They need Tik Tok, wow. I



01:00

know. I can't believe my heart. I can't believe my heart. My God

J

James Fink 01:22

and improvised lyrics there as well. By the way, the song she made up, I guess she told Laura and I, she surprised us. We went to, went to a farm yesterday where they, like Santa and Mrs. Claus every year there and like, on her lap. And then you take, like, a hay ride around the farm, and it's beautiful, because, like, snow everywhere yesterday, and they have hot chocolate, so it's really fun. And you see six, she just turned six in October, and, you know, she's in kindergarten, so like, I didn't talk question Santa until I was like, 989, or going to the farm. And she's like, Yeah, we're gonna get to see fake Santa today. And Laura, like, we just didn't expect that. Oh, fake Santa, what do you mean? She's like, Yeah, the real Santa, he's in the sky, like, this, like, pretty much, like, every Santa we see out in person is fake. And I was kind of, like, open to, like, hearing her reasoning. I was like, Oh, well, maybe it's and then Laura's like, Oh, how about that? But, you know, I wasn't sure what to do. So I sort of, like, was supporting her skepticism. Like, okay, maybe some of these Santas aren't real. She's like, me, if they're, I'll see if there's a string attached to his beard. I was like, damn, where's this kid getting this stuff from? Like, what? I don't think I ever thought about Santa string beard when I was a kid twice, like to have something in a cartoon. That's why I thought, that's why I was thinking like that. Or like, maybe her classmates have older siblings who are, like, introducing those things. Anyways, a sad moment, I was like, this might be coming to soon to an end, sooner than I thought.

S

Steven Ryan 03:01

Yeah, my I'm the youngest, and my brothers and sisters talk sometimes about how, like, like, getting to an age where, like, they weren't, like, they didn't believe in that anymore, but like, then the fun part is, got a younger step later your parents are, like, all right, part of your job is to, like, keep the keep the dream alive for your little brother until Easter. Yeah, exactly, yeah, you're in on it.

J

James Fink 03:21

Yes, that's right, exactly. That's so funny. That's probably how it will go for for my kids, I guess. So it's just us. Is I'm saying in the Plex email out Steve about the meeting, yeah, schedule a full team DB for Plex post GP and we working on Our portal, as we've learned.

J

James Fink 04:39

That's she will be here in 1015, all right, then we can get, sorry, Bucky, one second tool share.



Steven Ryan 05:03

Do you still want me to send the respond to David's email?



James Fink 05:05

Yeah, okay, I can. I can leave mine in drafts until you're all set. I just want to give some so, like, it's like, not till late in January now, but it's like, well, jpm is on the 12th, 13th, 14th, so you can't do it then still, actually got to write. We had to, like, write the report. So, like, I'm just, I want to provide some of that, not like, Hey, we're pushing this back as fucking late as we can. Does that make sense? So you still do what you gotta do. I'm just providing some context. So I schedule its full team debrief for plugs into a post jpm. We have been working on a report that outlines what we've learned, small molecule, structures, deconvolution, transcriptomics, etc. Few us are out of office essentially for the holidays, so expect to have a version of the full report ready for January, 6, seventh, after



05:54

chance to review with Steve and Luis.



James Fink 06:06

We will share the fort with The full team ahead of



06:42

1213, 1213, 1412, through 15th. All



James Fink 06:54

right, let me know when you send that, and I will. I will send this once it's all done.



07:16

James, James, okay,

 James Fink 07:25
okay, so let's talk about this experiment. This won't take the full hour, I don't think, but let's see what you guys think when we call Our powerful brains together. Steve Caitlin money,

 James Fink 08:01
okay, so

 James Fink 08:19
I wanted to talk a bit about the design for our first coal culture for ipsps and human so we want IP SPEs In human

 08:34
coal cultures.

 James Fink 08:40
The one one experiment is to mix

 08:45
ZX plus ng2, then synaptic assay.

 James Fink 08:57
This all cells called pre post, one option, the other one, which is only done more often,

 09:09
is dx, equals pre,

 James Fink 09:13
two equals post, and only balance previously as

 Speaker 1 09:23
apathy versus no kapazine.

 James Fink 09:31
Okay? And then from that data, okay, it's been mining ipsps, right? And at least the last time we heard we'll know more tomorrow. It's like six fobs equals 13 IPS fees,

 09:47
then plus GABA Zinn equals zero.

 James Fink 09:52
No, he's that criteria that when he does more fobs or some that's not bad. That's fine. So I think some of them probably spikes that have AHPS, they're getting classified as as a gap. So my my extrapolation of that is the guess, like, maybe, if you analyze 20 fops or 100 fobs, you get 100 ipsps with some average waveform, and then gap Z, and you have a couple and they're like, they're just clearly different, or whatever. So I think, I think we'll want to do number two. Guess I didn't do a good job indicating that this is the one we normally do. But I want to give you be clear that one's an option for us. So I want to talk about today is like, what are all the things we want to explore in this round to start answering questions about what's driving the lack of ipsps. Is it ngn two? Is it GABA? Is it the Deluxe? Is it chloride reversal, or is it GABA?

 10:53
Release those types of questions.

J James Fink 11:02

So let me, let me go list some of the things I'm thinking that we could do, and then you guys go from there. Yep. So one thing is try, and there's work that Kathy and Luis are doing separately for the inhibitory neuron parts. But like, can we make inhibitory neurons better, right? But let's assume that's a different question. I'm working on this AIM Under assumption that it's not the inhibitory cells themselves, it's the assay, right? Something else about right? So let me here's some ideas. We could do ACL dose response to modulate reversal. So I'm imagining is like under baseline, right Steve,

J James Fink 11:56

equal, shunting. Right answer like, ACL equals EPSPs, KCL equals Yeah. You'd want to do some experiment that's Yeah modulating, and you see like, oh, like, the number and size changes as we modulate KCL, and wish,

S Steven Ryan 12:23

yeah, there should be a way, like, like, a very easy version of that is, like, drop the resting membrane potential, the minus 80, yep, and you should get a bunch of ESPs. They should all go away with Galaxy, like, if he had PSP, like, if I was right that it was all chantane. And they were like, weirdly,

J James Fink 12:41

yeah, I'm just trying to think, would they be, would they be in him? I thought it'd be inhibitory at my SA, D, oh, no, you're saying there, if

S Steven Ryan 12:47

the Russian membrane potential is down, and the chloride, like, if you're

J James Fink 12:53

so if you, if you, if you add potassium chloride, you'll depolarize the neurons. Yeah, I guess we'll have to think of it carefully. So let's not define that's fine. So you get the point. Yes, we'll move it in two different directions, but and more in a more fine way. Yeah? So especially, I expect, like, probably two, like two doses here, one and two, so it's like a five point, yeah, sort of concentration response curve. And the goal is to have one condition, which we expect very big negative, right? One, which we expect some smaller negatives, one where we expect largely shunting, and then two that goes stepwise in the opposite direction, right? So this is like a patch up experiment where you do different holding potentials. We're not doing holding potentials or just changing the

S Steven Ryan 13:38

potassium chloride the there is a control that might be worth doing alongside that with the same culture where you do the same potassium chloride concentrations in the excitability assay, because there is a risk that as you elevate the resting membrane potential, those cells sort of like if you're doing too much, and they sort of go into depolarization block, or they're harder to activate. You won't see PSPs, but it's because you're not able to stimulate your presynaptic cells to fire.

J James Fink 14:08

Okay, so sister plate missing sensibility, just to see how that correlates with firing. Yeah.

i 14:23

Okay, so that's one. The other one is we could do

J James Fink 14:33

that by Pams to see if we can just boost. It's like it's taking sick Hong Kong signal, which is like, maybe the EPSPs get even larger, there's more of them, and then the true ipsp's get larger, and there's more of them, and more of them, just because the ones that are there sort of under, you know, largely undetectable, or seeing within the noise threshold. So you still expect that way to change. It would be a sign that, like, okay, these are really releasing. GABA, yeah,

S

Steven Ryan 15:07

we, I know we have done, sorry, you tell me that this is related part of the conversation, but I'm just sort of going in other directions. Yeah, this

J

James Fink 15:16

is just a brainstorming session. We're not gonna, we're not gonna. So we have a homeostatic plasticity experiment that Kathy sent around, yep. So give it a look this week. We're going to try and plate that next week, right, Kathy. We're also going to talk with Ben about the plastic and ordering and like designing, I think in two days. Think you're also part of that meeting about timing and treatment of drugs. So that would be another one to even plate next week or during break. So this experiment will not there's just too much other stuff to do. So like this one can wait. But the idea is to get the ball rolling so that we can send the design form next week and we can start thinking about what tweaks to make, and then played it when we get back.

S

Steven Ryan 15:53

We've probably done PCR looking for and I'm sure we've looked at this in the bulk RNA, see, but like looking for synaptotagmin, but yeah, basically like the different pieces of synaptic machinery to make sure that those were expressed. Have we ever done like, either ICC or western blots for like, basically like in our cultures to see, like, can we see pre synaptic, postsynaptic terminals? Are they co localizing? So you haven't done that, but we can do they're like auxiliary experiment, which is

J

James Fink 16:32

synaptic machinery, I think too hard to do this, like, just like the time it would take to optimize, like, closely looking at co localization, if snap tags, some of these would take longer than it would to actually just you, I mean, so I'm thinking of QCR to do this. Do these subcultures express these things?

S

Steven Ryan 16:57

Yeah, you have a better sense of the optimization than I do. I guess I worry that if we don't measure something directly, yeah, depending on what our outputs are like, if we end up with a bunch of functional results that are harder to interpret, then we're back where we started.

J James Fink 17:15

That would be the that's telling, though, but yeah, but so so I'm saying it's this. I'm just saying the staining part is this, the QB, sorry, can run in a week. Just get the probes, let this month. Like it's quick. Staining is like, first, I can still grow the neurons 30 days to you only have certain wells you can fix and stain. So it's RNA that stays forever, and you keep running it and, like, checking it, and it's scalable, so the plates was, like, right? And then you'd want just a bunch of different things, and then they're not going to work this time, like, off the bat. So they can have to do some plates you hold on the side that after you've, like, covered all the CO localization, and maybe snap the tagment works. But the antibody for this other one's bad. It just, it becomes, it's, it's unruly, that's all. So let's okay. We have to do some static quantification later, so maybe we can. But we want to do snap attack wind,

S Steven Ryan 18:17

like pre synaptic, yeah. We should look at, like, whatever the vesicular transporters are

 18:23

post to be like, jefferyn,

J James Fink 18:30

okay, that's a good idea. So let's think about those probes I want to order. Yeah, Luis has asked this question too. Like, do they have jeffrin or David? Like, are they? Yes, that's right. And like, if those are expressed, I feel good that they're likely, yeah, it would be unlikely

S Steven Ryan 18:45

for them to be expressed and not be in position, right? So that's fair.

J James Fink 18:48

But where I'm going with this is, like, just in terms of my comfort is I feel good. Like the data hunting show is pretty compelling to me, that, like, it's not zero. There are EPSPs there. They're not as many as I would think, right? There are ipSPs there, more than I thought. I guess we'll get maybe more from this, from Wednesday, whether those go away, if there's fewer of those, yeah, we'll find out. Yeah, they, I mean, I'd be concerned they don't, because there's no other transmission available.



Steven Ryan 19:18

Yeah, yeah. I mean, we know the Creed can be a little leaky, but,



James Fink 19:21

yeah, but, but we beat them. We played them in different days. So, like, highly unlikely to be. Like, the dose of Cree is high, like, 300 nanometers high, whereas synaptic virus, we normally give 20 nanometers, or was it? Is it 40? It's a low amount. Like, yeah, it's very small magnitude, yeah. And then we did only the presence of the one cell type. So it feels like, in those conditions, we don't for synaptic acid. In the publication figures, that dose makes only Quasar expressed. This is in rodent cultures. We have both, right? So it's like if you play those cells and you add three engine out of these Cree and then wash it away and give both Quasar and Sheriff. It does not express any Sheriff only quasar. So it's on the tail end of, like, it should be very strong. Okay, so anyway, that would be one, and I think this would be another, just like, can we modulate this small number? It's like, our baseline. It's like, oh, there's some here. Can we get that to go up or down with either Pams or this another one would be to do singles will



20:27

stem, S, C, S, using functional masks.



20:40

Label, DLx



Speaker 1 20:43

with JFP, RFP, stem with different durations plus strength,



21:01

record, PSPs.

 James Fink 21:05

So we've done this in the past, but we are doing the blobs, not functional two. This is not a single cell experiment where we need to do 500 cells stimulated. We take a picture, find three cells for DLx, do a stim. Find if any of those three cells fire. If they do fire, just stimulate that cell and see if we get an IPSP or an EPSP or nothing, I feel like that could be telling. Can imagine doing that in the context

 21:37

of and ACL?

 James Fink 21:41

Dose, concentration series. You can imagine that you're getting some set of EPSPs, IPSC under this condition, and then goes out and down there. Why I want that one is because I want to see the inhibitory neurons fire, and I want to see how that differs versus the excitatory neurons firing. And why I say that is because one of my concerns, and Hank talked to me about this, about those weird artifacts, right? He thought the stem artifacts for like, there's a spiking versus non spiking. It's like, it's weird. So I want to make sure that, like those STEM artifacts we see in inhibitory neurons, amazing. There's next step for Hong Kong are not how STEM artifacts look like in excitatory neurons.

 Steven Ryan 22:37

Interesting. Yeah, and it's a little unclear to me, historically, what the stem artifacts actually are.

 22:44

Like, physiologically, yeah,

 Steven Ryan 22:49

like, like, I often don't question them, because I'm so used to like, yeah, when you do electrical stimulation, of course, you get a stem artifact. But like, optically, we really shouldn't have one, right? Unless it's crossed off. But yeah,

J James Fink 23:01

I think it has always been thought to be crosstalk, at least talk about it, but I haven't explored it.

S Steven Ryan 23:07

Yeah, I think the two thoughts here, one is, you haven't thought about doing this mapic assay, where the where you you actually have. So I guess we currently have two different constructs. We have a sheriff and a quasar. The other way you could do that is your postsynaptic cells get Quasar in your presynaptic that's so that's this post synaptic Creon, yes. So you have a Creon Quasar in a Creole off Sheriff P to a quasar. So your presynaptic cells have Quasar as well. So you can see the fire.

J James Fink 23:43

Do we have villenty like that? Can you take that as an action item,

S Steven Ryan 23:49

but like, be a thing for us to think about, for the asset that's a good that's, yeah, I like the single felt, then it does have the

J James Fink 23:57

well, so yeah, that's where I'm going, which is so I've always suspected that maybe, like, I'm just again, odd that, like, we're not seeing more EPSPs, yeah, if it's truly a KCC, one, nKC, one KC, two thing. So I thought like, oh, maybe the gamma cells are just not responding. And this is the Hong Kong's point. Like, well, the first and the second stem are these little EPSPs, and they also reverse. So I don't know if there's some, like, interesting thing going on there. So I thought like, maybe, well, we know the inhibitory neurons fire, but, you know, they fire less because of the DFB data. There's not as many sources. So certainly, like, we're not getting, it could be that, like, we're actually not getting robust firing in all those cells as well as we thought we were. Have we

S

Steven Ryan 24:41

tried just expressing the regular, I was gonna say, is put the excitability constructs in just DLx neurons, and then run the synaptic protocol so you so they all express Quasar and Sheriff. It's not exactly apples to apples, because you don't know if they're expressing the same I'm sure if,

J

James Fink 25:02

yeah, I was just thinking like, if you take in the same culture, since you have green and red, and we'll have many more excitatory neurons and inhibitory neurons, and we do single cell stem, and we say these are 7789 it's like we know the cell. We know DLx can fire from our previous data rounds. We know ng two can fire from our previous data rounds. The DLx during JF, P gives a change, relax the filters. Does some of that have to do with their ability to respond as well? So we know their quakes are might be a little different or dim or whatever, right? But like are, they also different in their engine to expression, or their their synapse, synapse and expression is different, so they thrive as much. So then you ask, all right, let me do single cell stem so I identify functional cells. Now I do a single cell stem with this power at this duration, how many spikes do I get from engine to across hundreds of cells, and how many spikes do I get from bloxel, right? Oh, they're 30% reduced in their firing. And that tells you a bit about what might be happening when you do actual synaptic assay, like, oh, we might need to up the duration of this to maximize what we normally do for engine to culture, like, if it requires double the power, or, like, double the duration. Yeah. So you might say, okay, for to match the NmE to firing, they might say, okay, during synaptic assay, we should double the blue stim so we get these cells to drive more.

S

Steven Ryan 26:33

Yes, that makes sense. I think the only risk with this is just that we, I don't know if we've solved the engineering problems for a single cell stem, yep, but so I would say, as a de risking like, if you take either, either in the same place, like, we just modify the single cell stem pre Protocol, or like a sister plate, you give them like, like, if you do it in the context of the single cell stem, we Do a ramp to find the Yep, do the synaptic? Add in some of the synaptic. Like, it's like 90 the watts per centimeter squared, whatever the duration is. Add some of those in because then if we have some sort of engineering problem with a single file them, we can still at least look at those pulses and see,

J

James Fink 27:16

right, right. Okay, okay, that's a good point. So you're saying, if you don't use a DMD, you're still just, you're still getting, giving a synaptic gap assay and post hockey, good luck. Did those cells fire? Yeah, okay, that's a good idea. I like that.



27:31

Yep, that's good. So,



James Fink 27:39

yeah, that's like excitability with different



Steven Ryan 27:41

Yeah, because we like that. Yeah,



James Fink 27:44

that's that's interesting. Okay, I like that.



27:58

What else could be challenging here,



James Fink 28:05

your GABA, gabazine, yes, no, and then the PAM should get you at the receptor question, yeah, like again, if you're not getting any PSPs in the presence of gabazine, even though you're getting few, it's still telling you the cells express

S

Steven Ryan 28:19

receptors just thinking about, like, physically, what are things that could be going on? This is a weird one, but I don't think we've ever mentioned this before. What if, like, postsynaptic cells express JCC to have a good chloride reversal potential, they have GABA receptors, and the pre synaptic cells are releasing GABA, and they, like, kind of have synapses, but some of the like, structural anchoring proteins are not expressed, either not the right types or not the right, like mature levels. And so there's just, like, too much space between the pre synaptic and postsynaptic term. You could get great presynaptic activity and have all the right post act machinery, but you'd get like, wimpy, gross little PS, IPS, right? I don't know what. I don't know if that's actually plausible. I don't know what genes mediated, but it would be really hard to detect, and it would look like you just don't get ipsps, yep, no, that's good point.



29:19

How would we you Yeah. How will we solve this?

S

Steven Ryan 29:26

Okay, what's that? What's that modality where people use like ramping electrode voltages to measure like serotonin and dopamine concentrations?

J

James Fink 29:39

It's like isometry, or somebody, it's

S

Steven Ryan 29:48

like dopamine. There's a voltammetry, voltage, neurons. We're just thinking,

J

James Fink 29:57

brainstorming different possible experiments. Kathy, are you following some? Are you learning some? Yeah, what do you what are your questions here?

 Speaker 2 30:06
No, me, I was just looking at what aims were,

 James Fink 30:09
okay, yeah, positive allosteric modulator, so it should make the GABA signal stronger, which means, if they're there, they're wimpy. They don't went into the perspective, yeah, they open, they open the receptors longer, so they should get more current flowing, which means you Get bigger IPSP or EPSP if they're depolarizing.

 James Fink 30:42
Let me speed all of this into otter or into tragedy. Otter

 30:54
home, meaning I

 Steven Ryan 31:11
there are, this is now going down kind of a rabbit hole of an assay development, but there are GABA sniffer sensors, like express a sensor and what to see if we're getting out of release. Guess.

 James Fink 31:26
Let's ask, let's add that to the list. Can you add that Kathy to your list of things to look up? James?

 Steven Ryan 31:40
When I were in grad school, there was a line of enzymes image optical glutamate sniffers. Then they're they're basically like, when they bind to the target glutamate or GABA, they light up. And so it's a way to make synaptic it's another way to make synaptic transmission into an optical assay,



32:01

but it's chemically specific instead of voltage.



James Fink 32:04

Yeah, so is it one way to assay whether we're actually getting GABA in the i



32:54

Let's See what they



James Fink 32:56

what they say about possible hypotheses here. I sent the whole conversation to i



James Fink 34:57

Let's see what it says here. Let's see how good German is. I'm just trying to think, because I feel like we've covered a good amount of possible experiments that we can venture into is like gap, glute, GABA Sniffer, over expression with lentivirus. Of cases. I mean, those are the next tier, while we're also like exploring the different, newer inhibitory neurons those matter.



James Fink 35:37

So first one shunting, second one weak signals below detection, which would be Pam's, would address that third one, missing synaptic machinery, right? Like you said, Too much physical space. Inhibitory neurons are not firing.



35:58

Get that from the single cell spin, yep,

 James Fink 36:01

depolarization block. If the resting potential is elevated, the self might enter deep polarization block, rendering them unable to fire. So that's an interesting question. They're like, sort of less mature, or something like that. Then, yeah, I wonder if that I'm just thinking like if that could explain some of the excitability. Like, in short, I talked about this, like cells are brighter or not. Like, imagine the cells are in depolarization block, like inhibitory neurons just haven't matured, right? What are their baseline fluorescence? Look like being quasar? Yeah, it would be brighter, brighter, right? Not dimmer. I'm just saying that almost like if they Yeah, like, if they have depolarized, yeah, 40 no holds, they would just be bright. There's be bright and they're not, they're dim. Well, was a feature that would distinguish them from the engine to that you had to relax in

 Steven Ryan 36:53

the well so they could be,

 36:55

it could be brighter

 Steven Ryan 37:00

per quasar, because the quasar is fluorescing more, but still dimmer on average, because there are fewer Quasar proteins expressed, and then that would so like, what we the thing we had before is that they had a lower SNR, basically, which, Like, which could be, could the lower Quasar expression?

 37:19

I uh, or could be something about maturity.

 Steven Ryan 37:26

The spikes are shorter and like thicker in immature neurons.

J James Fink 37:31

But are they that way in Deluxe from that data set, do you remember? I don't remember. We should look. We should add as an action, aim for like one of the next aim, one Ruggedization check ins to go review some of the historical data, the DFP, DLx data, in terms of what, what's making the DLx cells have lower sources. Just we should dive in, like, this is like the type of stuff, like with funkangs, which is to say, like, okay, they're they're brighter, but their SNR is worse, or their spikes look like this. And we can start building an F building evidence to say maybe they are depolarized compared to engine. May just haven't matured to really so that, which actually when you get blue stimulation, this actually would happen right? If they are closer to threshold, their actions are these sort of shorter, broader, they would get likely less. Release of GABA from the terminals, they will get wimpy, yeah, release of

 38:24

Yeah. So here's

J James Fink 38:28

some other pre symptom mechanism, lack of Gad, enzyme activity. They may be immature and fail to express gad. I think we know that they do ve gat expression failure if GABA is synthesized, the facility to GABA transporter might be absent or not functional. Vesicular fusion failure, axonal conduction failure, high expression of sheriff in the axons, combined with strong blue light stimulation might clamp the membrane potential of the axon, preventing the propagation of the action printer to the terminal? Yeah, I'm trying to picture that my head

S Steven Ryan 39:05

well, because the axon, the axons require like propagation, but so if you stimulate part of the axon at the same time as the cell body, then then those sodium channels are in, are inactivation, and it can't propagate, you get a interesting,

J James Fink 39:22

insufficient axonal harborization and DX neurons may be healthy, but fail to extend axons far enough or branch sufficiently to physically contact engine two neurons, glutamine starvation, the culture media may lack sufficient glutamine, a precursor for glutamate and subsequently, GABA lean to transport. That's probably unlikely, since we



39:40

get glutamate, fine



James Fink 39:43

synaptic cleft and neurotransmitter overactive GABA uptake. GABA transporters on the neurons or surrounding glia might be clearing, release GABA from the cleft so rapidly that never reaches a concentration sufficient to activate that's unlikely. Receptor desensitization, high ambient levels of GABA in the media tonic GABA may have caused the postsynaptic GABA a receptors to enter desensitized, closed state, rendering them unresponsive to phasic release. GABA B dominance, the NGT neurons might predominantly Express metabotropic GABA B receptors, rather than ionotropic GABA a receptors, GABA B responses are slow, hundreds of milliseconds. Might be filtered out by your data analysis. Can you guys make a note to check the Opto, the X seek viz for GABA a versus GABA B in the engine two if the cold culture contains aim of cells that may be aggressively unlikely All right, post synaptic mechanisms, diffuse receptor diffusion, GAVI receptors may be expressed, but not clustered as synapses. Extra synaptic receptors generate smaller, slower, tonic currents that are difficult to detect. Intracellular poly poly amine block, if the engine two neurons Express calcium permeable AMPA receptors for specific channel subtypes, endogenous intracellular polyamines can be blocking the pores, phenomenon that can sometimes affect chloride channels depending on subunit composition, outward rectification. The specific subunit composition of the GABA receptors expressed may result in strong outward rectification. Mean they pass current efficiently only at potentials far from where the cell sits during the assay. Immature chloride, excuse me, while hunting was discussed a related but distinct issue of size cell chloride washout. The cells are physiologically compromised. Chloride internal may equvalate with external media eliminate the chemical driving force, pH sensitivity, optical stimulation and high metabolic activity can transiently acidify the local extracellular environment. Cave a receptors are inhibited by low pH, which could dampen the current.



James Fink 41:58

Optical stimulation and high metabolic activity can transiently acidify the local extracellular environment. GABA, a receptors are inhibited by low pH, which could dampen current we should look into that, mostly in terms of like, what the pH of our buffer is.



Steven Ryan 42:16

Yeah. I mean, I assume we imaging buffer. That must get pH when it's made. I guess you don't make that, even though,



42:29

yeah, pause gently check pH of that. Here's



James Fink 42:32

some other ones. I'm just gonna read them for the sake of completeness, spectral crosstalk, pre activation, if the red light used to image Quasar as a spectral tail that overlaps slightly with Sheriff DLx. Neurons might be chronically activated during the baseline period before



42:48

blue stimulus. I'm



James Fink 42:51

just trying to think of that would that we wouldn't capture with just the engine two cases, right? Which I believe that that was true. It would be true for ngn two as well. Inhibitory currents can be slower than excitatory ones. If the frame rate of the quakes are imaging is too low, or if the indicators filter sensitivity is non linear in the hyper polarizing direction, it may fail to resolve the specific wave form of IPSP by unlikely. That's right, high pass filtering, if the data processing pipeline uses a high pass filter to remove photobleaching trends, it might inadvertently subtract out the slower, broader hyperpolarization of an IPSP, leaving a flat line. Yeah.



Steven Ryan 43:30

I mean, that's the thing that I worry about generally with the synaptic assays, because they're so much slower than spikes. But again, they come out nicely in rodent neurons. That's the same pipeline,



James Fink 43:39

so, right, but they're bigger. So do you think that they would be just

 Steven Ryan 43:44

that there? There's a little bit more signal there than there appears to be, but it's being squished by police correction. Yeah.

 James Fink 43:52

So, like, so like, in rodent, you still get robust ipsp's. Because they're they're large enough size. Nothing filtered out. Yeah, yeah. Are the traces of Hong Kong looking at bleach correctly?

 Steven Ryan 44:04

They probably are. But we should ask them to check,

 James Fink 44:07

is there any process by which there's police correction, and then, actually, when it does a correction, it flips the IPSP to be an EPSP, because it's like normalizing in a weird way,

 44:16

I don't know. Like

 James Fink 44:19

you can imagine there's a scenario with, like, the bleach curve, and so you, like, correct it equally on each three stems, but for the first two stems, it winds up being corrected positive. And then the last one, it's,

 Steven Ryan 44:30

yeah, oh, I see what you're saying.



James Fink 44:33

Yeah. Because they're so small, that's the only reason I bring you up. Like, if they weren't so small, maybe it wouldn't, it would be. So that's something we should just check. I think we would



Steven Ryan 44:42

probably have to rerun the movie analysis. Okay, and genes pipeline doesn't save the non police corrected.



James Fink 44:50

Okay, all right, let's, let's take that as a note to discuss tomorrow, like me. Is the little project we can give Hong Kang you



Steven Ryan 44:59

want to give young fit, there are chloride sensitive dyes, so we can measure, we can estimate chloride concentration intracellularly in different cell populations, but we would probably need new optics and scopes.



James Fink 45:16

That's hysterical our rodent neurons.



Speaker 1 45:20

Beautiful. NPS piece, large and frequent



45:33

NmE. It's this change. Your Are you



45:55

learning something? Kathy, what you wish for you? Looking? At sequence,



Speaker 1 46:05

looks like this expression of GABA A



James Fink 46:08

is higher than B. There's no B, sorry,



46:14

P, so GABA rep P is



46:20

associated protein. So as such, I'm not able to find the GABA



James Fink 46:29

R A, GABA R A, it's not all GABA R A, it's G A, B R a, it's like weird. So you should look up make sure you have the right gene name.



Steven Ryan 46:40

Yeah, take A, E, R, A, yeah, those are the Alpha subunits of the different Gavin over subcomers.



46:52

Look at our a, one, no, a two is definitely in there.



James Fink 47:05

Okay, so here, here's his, here's its, revised hypotheses based on our other assays. So hypotheses are now unlikely and ruled out Quasar kinetic sensitivity, since running neurons show large ipsp indicators clearly capable of capturing inhibitory events, analysis, filtering artifacts, the same pipelines, sees, wrote in ipsp is some filtering human ones, unless human ipsp are vastly



47:30

slower. Could be a second time for this.



47:33

Er, one or two. I



Steven Ryan 47:49

There's a few the office of units of GABA, a that we express mostly a alpha three,



James Fink 47:56

okay, yeah, login. You're saying, Yeah, I'm here. I am. I didn't realize I was logged in.



Steven Ryan 48:07

Alpha two and five are in a low level. Three is at a pretty high level. And then I don't remember. I think I would be doesn't have sub unit the same way, because it's broken coupled. But so then type one and type two are



Speaker 1 48:23

expressed very intentionally. Yes, we have compared to sin three. I think, yeah, this



48:34

is an energy into neurons. I



James Fink 48:49

here's a couple other things that says GABA switch hasn't occurred. So that's good, right? That's one option. BLX, presynaptic immaturity, synaptic ZIP Code mismatch, neurologens neurons, we now know NGT neurons can form synapses EPSPs that are present. The failure is specific to inhibitory neurons. Inhibitory synapses are very specific adhesion molecules. Neuroligin Two and gephrin,



James Fink 49:18

yeah, n, e, u, r, o, l, i, G, i n2,



49:33

okay, kind of what I was thinking of earlier.



James Fink 49:39

This is interesting. N, G, N, T. Neurons are generally generated via force transcription factor protocol. It pushes them rapidly toward excitatory fate. They may be transcriptionally prioritizing aim and MBA receptor trafficking to the membrane while lagging and GABA or separate trafficking.



Steven Ryan 49:59

So how would we so that's sort of getting to that, like PCR versus, yeah,



50:05

that would be harder. Yeah, exactly. Wonder if there is



Steven Ryan 50:15

a vendor who will do ICC for GABR receptors and for all these other things like



James Fink 50:21

this says there's a smoking gun here in the expression data. I shared the screenshot of acharya's data. You ready here the missing fast switch from GABRA a one versus GABRA a three. The most glaring signal here is the absence of GABRA A one and the dominance of GABRA a three in the developing brain. GABRA a three. Alpha three is the embryonic fetal slip unit. It generally produces currents with slower kinetics in the adult GABRA One the rodent comparison, your rodent neurons likely Express high levels of GABRA a one responsible for the fast, sharp, large ipsp's Your engine neurons are expressing the field subunit, the resulting ipsp's are likely broader, slower and potentially smaller in amplitude. If your optical analysis pipeline or your mental model is looking for sharp, fast transient, you might be filtering out the slower events as drift,



51:16

the silent giant GABR,



James Fink 51:18

gab br One, the expression of GABR B, GABR one, GABR B receptor one is massive, 200 TPM dwarfing the ionotropic channel subunits. The GABA B receptors are metabotropic. They do not open chloride channels directly. Instead, they trigger second messenger cascades. They eventually open potassium girk channels. GABA B responses are slow, hundreds of milliseconds to seconds. They do not look like spikes or standard ipsp's. Quasars voltage scan, if gabrab One is this high, your cells might be under strong tonic inhibition and clamping, mempotent and making it even hard to see distinct synaptic events. That also might be why they don't respond as much to Quasar if they're clamped, because they have such high levels of that



Steven Ryan 52:04

is interesting, though, because if GABA B is, like, tonically highly active, they should have a really hyperpolarized membrane potential, yeah?

 James Fink 52:14

Well, like, maybe compared to their normal baseline, which we don't know what that clamping them, but that's a yeah, here's more tonic inhibition GABRA Five. GABR five is classically located, extra synaptically Outside the synapses, and mediates tonic inhibition, rather than phasic inhibition. If the pre snapping DX stars releasing gabap, it may be spilling out and activating GABRA, a sort of secret and noisy clamp on the cell, rather than discrete events.

 James Fink 52:46

So here's a couple to recommend. Data, GABA B antagonist, CGP, 55845, the noise decreases, or the cell becomes more excitable. You know, these receptors are active force the signal. Since GABA three is a dominant channel. Look for a modular specific to Alpha three, or general benzodiazepine, benzodiazepine with diazepam, which affects alpha two, three and five. This might boost the wimpy and brown occurrence enough to be seen. Does you small guys are so good thinking now, one last little hat trick. Hong Kang sent me his IPS protection. I want to, I want to feed it the the data. Let's see if

 53:41

it does anything with this. Here is some data to consider. I'm

 James Fink 54:26

How did this go? I just had it on my computer, but not including uploaded.

 54:30

I don't see it recent desktop.

 55:04

Me. Oh, here we go. Okay,

 James Fink 55:21

let's See what it says about my data.

 James Fink 56:19

Okay, so here's what it says. Ready? You guys ready? Yeah, slide deck proves the biology is actually working, but the detection was a bottleneck. It's not zero inhibition. New Approach detected 13 ipsp's in the treated culture versus zero in the gabazine treated condition. This result is critical. Gabazine control worked. If there are truly no no PSPs, GABA seems we've changed nothing. The fact that signals disappear with GABA seems provides that GABA is being released and is active. Is active being receptive. The artifact drifts, confusion. The traces show significant drift or slow waveforms in the rodent cultures, ipsp's are sharp negative deflections in the human cold culture traces identify ipsp's are broader and sometimes barely distinguishable from noisy baseline without the five sigma thresholding, the recovery 800 units notation associated venture long lasting, consistent with the GABBR a, three, GABBR B, r1, slow connects hypothesis.

 57:13

That's interesting.

 Steven Ryan 57:17

Yeah, we should run some of this past time trying in terms of this analysis,

 James Fink 57:22

it says analysis permanently adopt the new approach criteria peak time of less than 150 milliseconds for human data, do not filter out low frequency events that might represent genuine GABRA Three kinetics. Acknowledge the GABRA a one is missing, expecting rodent like sharp IPS pieces biologically unrealistic for the specific developmental stage

 57:43

of engine two neurons. So

 James Fink 57:44

another thing that I would check Aishwarya is ordering the probes for these things, and then we can test them in the v1 versus v2 versus v3

S

Steven Ryan 57:52

should also test them in the rodent neurons, just to confirm that the because these are still pretty juvenile rodent neurons that we culture. So It surprised me if they haven't made that transition either.

J

James Fink 58:02

That's interesting, right? Okay, that's good to know. All right, so I think that's we have three minutes, so I think that's sufficient for next steps. We have some thinking about things we can change, and this, this will address the biology part, but we'll address some of the concerns we have, and we'll provide answers. Here's our action item for you, ashwora or Cathy, whoever hung KN has been analyzing one of the data sets. I can't remember what the round was. Can you find the design form for the round he's been currently analyzing? In addition, I want to find the round where we tested v3 for this. I'm pretty sure. So there's a concept condition see that have drastically changed the excitatory neuron profiles, right? We know synapses goes up. We know that NmE, receptor units go up, amp or receptor subunits go up, like, it's a nice, beautiful dose response. That's where it has the RNA. Yeah, Troy has RNA from that we're checking it for. I think Aishwarya did stuff for like five HD to a and KCC. So we have some stuff. I would also be curious Aishwarya, if, if we have the functional data we saw Hong Kang when he does all the FOVS for this experiment, the one he's been working on, GABR, Z knot, if he does it for all the v3 say, hey, we did the inhibitory acid. I think we did this. We did the inhibitory assay in the presence of, yeah, I'm pretty sure we did. I'd be curious to know if it also goes up, you know, if the number of ipsp's goes up, it would be good. It'd be good. I mean, it doubles or something. That's a huge win.

S

Steven Ryan 59:32

Yeah, yeah. I mean, I said yeah, that we already have a protocol that's right, moving in the right direction, yep.

J

James Fink 59:38

Okay, are those good, good takeaways for the meeting. Yeah, you want, you want to be in charge Kathy of taking the other conversation and summarizing into a debrief.

A

Aishwarya Dhandapani 59:50

Okay, thank you guys.



James Fink 59:55

Sure of this cool. It was cool. Never easy, though, let's never make it easy for you. Never.



Aishwarya Dhandapani 1:00:12

Thanks Kathy, absolutely, anytime. Yeah, important question. We'll see what kind of snacks we have.



James Fink 1:00:25

They're probably not in the kitchen, but there's good amounts of the closet. There's some pistachios, there's some chips.



1:00:33

Talk to you soon. Bye.



1:00:49

One second a short way out. Make use the bathroom. Quick. Okay, sure you



James Fink 1:02:40

i Okay, Aishwarya, what's on the agenda? I forget what we're gonna even talk about in this meeting. I know we, I asked you to have another meeting.



Speaker 3 1:03:03

Yeah, we were going to talk about the vacation, which I could, I would have time to tell today. Then you want to push that. Yeah, let's push that. But I do have a question for you this. I mean, this meeting, this, this one, like this one, this one. When we did any CP two DLx



1:03:25

phenotyping, we did excitatory and synaptic



1:03:28

assays. Yeah.



1:03:32

Do we have good synaptic data from that one or no?



James Fink 1:03:35

No, because it was excitatory synaptic transmission.



1:03:40

Okay, the inhibitory neurons, children, blood GABA.



James Fink 1:03:45

So inhibitory neurons can receive input, right? So we were just looking at EPSPs onto those cells that wouldn't tell us much about them. I don't think it's a good thinking. I mean, one thing to think is, this might be more of a what if that's just a lemon, a genetic background? Yeah, mature. They only express low levels of gap. You know, I'm saying that could be a genetic background thing. So that doesn't mean we should think about this experiment, More than one genetic background



1:04:29

that makes sense. How far



Speaker 1 1:04:43

half hour so I can work for that. Okay, let's just work. How's the sun? So I did this. JF, I'm



1:05:04

think i.